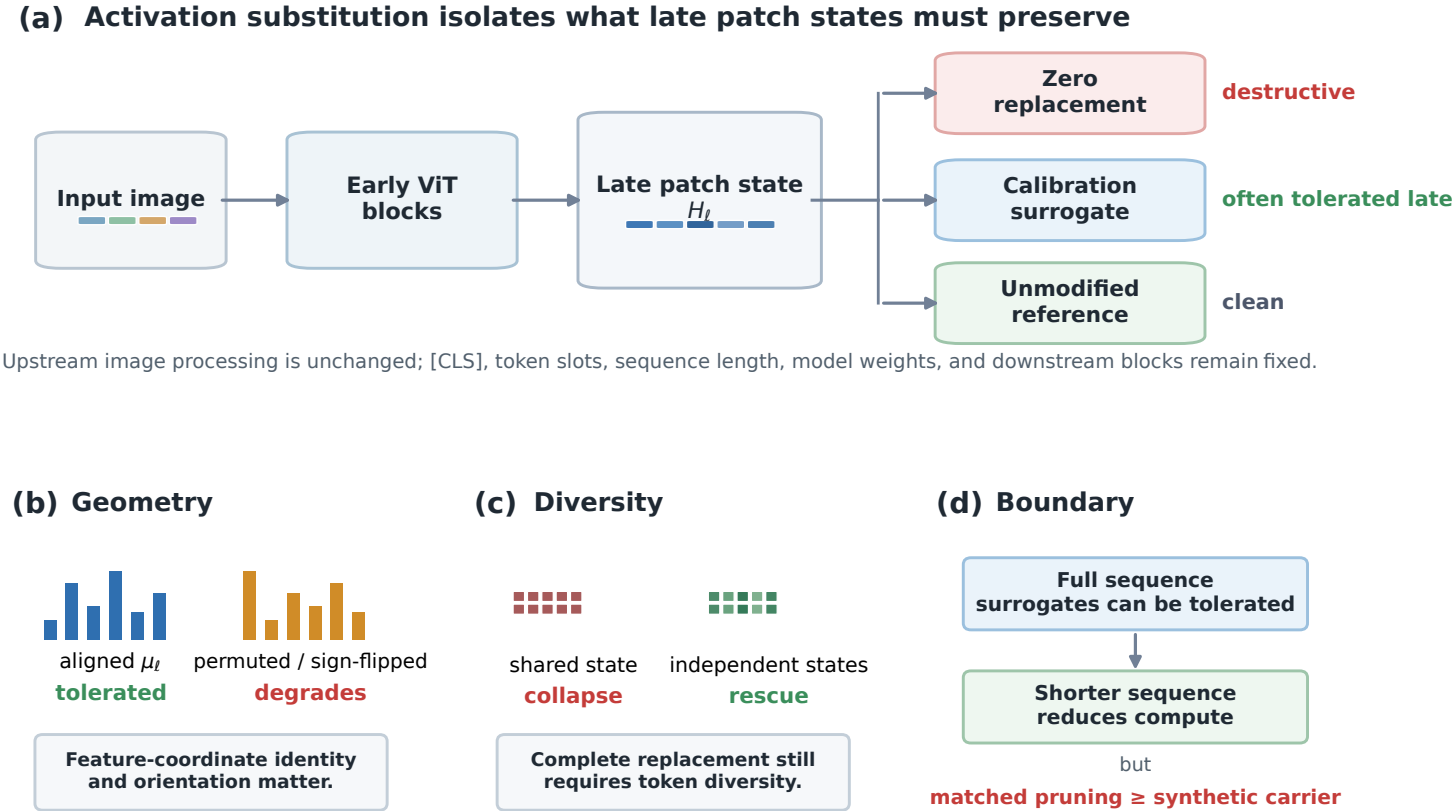
(a) Activation substitution isolates what late patch states must preserve
Input image
Early ViT blocks
Late patch state
H_ℓ
Zero replacement
destructive
Calibration surrogate
often tolerated late
Unmodified reference
clean
Upstream image processing is unchanged; [CLS], token slots, sequence length, model weights, and downstream blocks remain fixed.
(b) Geometry
aligned μ_ℓ
tolerated
permuted / sign-flipped
degrades
Feature-coordinate identity and orientation matter.
(c) Diversity
shared state
collapse
independent states
rescue
Complete replacement still requires token diversity.
(d) Boundary
Full sequence surrogates can be tolerated
Shorter sequence reduces compute
but
matched pruning ≥ synthetic carrier